MUHAMMAD NABIL RINALDI

AI & Backend Engineer · Jakarta, Indonesia · 26/02/2002
mnabilrinaldi@gmail.com · nabilrobert.my.id
github.com/NabilRobert · linkedin.com/in/muhammad-rinaldi-129111239

Latest as of October 2026

SUMMARY

Product Manager and Full-Stack AI Engineer at an AI software startup, with a Bachelor's degree in Economics from Binus University. Experienced in AI application development, agentic AI systems, data analysis and web development, with a focus on reliability. Built and run a live database MCP server (163 tests, 16 of 16 red-team attacks refused), an evaluation harness that catches agent regressions in CI, and a security gateway that cut scripted attack success from 12/12 to 5/12. Reported a bug in the UK AI Security Institute's Inspect AI; it was accepted and my fix is under review. A hard worker and always curious, I learn whatever skill the work needs.

TECHNICAL SKILLS

- Languages: Python, TypeScript, SQL
- Backend & data: FastAPI, Node.js/Express, PostgreSQL, SQLAlchemy, Alembic, Prisma
- AI & agents: MCP, multi-agent pipelines, LLM evaluation (Inspect AI), PydanticAI, prompt-injection defence
- Frontend: Vue 3, Nuxt 3, React, Tailwind CSS
- Infrastructure: Docker, Linux, Nginx/Caddy, Cloudflare, GitHub Actions, OpenTelemetry, VictoriaMetrics
- Testing: pytest, Vitest, pyright, ruff, mutation testing
- Spoken: English (IELTS 7.5), Indonesian (native), Japanese (NAT-TEST N4)

PROFESSIONAL EXPERIENCE

Beobot (formerly Edge Robotics, an Elemes Group venture) **January 2026 - Present**
Product Manager & Full-Stack AI Engineer **Tangerang Selatan, Indonesia**

- Own product and engineering for AI integrations: map clients' workflows, write the specs, and build the systems end to end with TypeScript, Nuxt/Vue, Node.js and Prisma.
- Built a multi-agent prototype for a car dealership (three writer agents and a critic; code decides approval). It ran unattended daily from August to October 2026: 65 of 68 daily runs succeeded, at a median model cost of $0.17 per run.
- Found that its quality metric understated failures about threefold (32.5% vs 95.0%) and replaced it with severity-based approval. Public case study, "Pitstop": pitcrew-public.my.id.
- Built a marketing and content orchestration platform. Use Claude and other AI coding tools for speed, and review all generated code before merging.

Elemes Group **September 2025 - May 2026**
Junior Frontend Web Developer **Tangerang Selatan, Indonesia**

- Built responsive interfaces from designers' mockups with Nuxt 3, Vue.js and Tailwind CSS, matched across screen sizes.
- Contributed to a POS ordering application: data flow and state passed between pages and components.
- Fixed bugs, refactored into reusable components, and checked each change against existing features to prevent regressions.

Dipo Star Finance **March 2025 - August 2025**
Associate Data Analyst, MMC Business Administration Dept. (Contract) **Jakarta, Indonesia**

- Analysed company data with SQL and Excel, and traced what was driving the figures.
- Analysed a regional dealer's performance (Pekanbaru) and presented the findings to support business decisions.
- Supported back-office operations for the Mitsubishi booth at GIIAS (GAIKINDO Indonesia International Auto Show).

Bank Indonesia (Central Bank of Indonesia) **August 2023 - January 2024**
Intern, SME Development & Consumer Protection Department **Jakarta, Indonesia**

- Data analysis and quality control of goods for the department's SME programmes, in an internship programme run by the Ministry of Education (Kemendikbud Ristek).

EDUCATION

- **Bina Nusantara University** — **Tangerang, Indonesia**
  Bachelor of Economics, Major International Business Management (Global Class) — **2020 - 2024**
  GPA 3.68

- **TAMA University** — **Yokohama, Japan**
  Bachelor of Arts, Major in Global Studies — **August 2022 - February 2023**
  GPA 4.0

SELECTED PROJECTS

MCP-backbone · read-only database MCP server · Python, FastAPI, PostgreSQL · live at mcp.my.id

- Every query is parsed, cost-checked, run in a read-only transaction and logged. It refused 16 of 16 red-team attacks, with 163 tests running in CI.
- Benchmark: gpt-5-mini answered 29 of 30 questions correctly through it under strict grading.

agent-eval-harness · regression gate for AI agents · Python, Inspect AI

- Fails CI when a change lowers agent quality by more than run-to-run noise. It caught a drop from 98.8% to 78.8% caused by a "save money" config change, with no false alarm on A/A checks.
- Publishes the three degradations it missed, alongside the ones it caught.

mcp-security-gateway · security proxy for MCP servers · Python, ONNX Runtime

- Adds auth, per-tool policy, a tamper-evident audit log and prompt-injection detection. It cut scripted attack success from 12/12 to 5/12.
- On a held-out test set, the detector caught 49.7% of injections at a 1.2% false-positive rate. The proxy adds about 3–9 ms per call.

Open source · Inspect AI (UK AI Security Institute)

- Found and reported a bug: tools with optional parameters failed with a 400 error under strict mode. The issue was accepted, and my fix is under review as PR #6184.

agent-observability · cross-project tracing · OpenTelemetry, Jaeger · deployed (private viewer)

- Follows one agent request through the gateway and MCP server to the database as a single trace (36 spans), adding 0.39 ms (1.4%) to the median tool call.

Pitstop · multi-agent ads analysis · TypeScript, Node.js, PostgreSQL · demo: pitcrew-public.my.id

- Three writer agents and a critic agent, with approval decided in code; 548 critic reviews over 231 outputs in a two-month daily run, and 388 backend tests.
- Public case study backed by evidence files, with its limits stated: synthetic ad data, and the live Google Ads path never ran on a real account.

In progress

- systems-monitoring: alerts, uptime probes and SLOs with error budgets on $0 tooling; 15 end-to-end tests, always-on stack capped at 368 MB of memory.
- Juku: an agentic JLPT study platform for Indonesian learners, built on the projects above (in development, not launched).

ACTIVITIES & INTERESTS

Nippon Club, Bina Nusantara University (2020–2024) · Rohis, Bina Nusantara University (2020–2024) · Operational committee, Masjid As-Sholihin Yokohama (2022–2023) · TAMA University basketball club (2022–2023)
Interests: AI, programming, e-business, marketing, business management, finance, and Japanese language and culture.